

REOS Energy Platform

Complete User Operating Manual & Technical System Specification — Version 2.0



REOS (Renewable Energy Operating System)

Complete User Operating Manual & Technical System Specification

TABLE OF CONTENTS

Chapter 1: Introduction & Core System Philosophy

- 1.1 Executive Vision & Product Scope - 1.2 Target Personas & Use Cases - 1.3 Key Innovations & Platform Values

Chapter 2: Architecture & Technical Foundations

- 2.1 Full-Stack Monorepo Structure - 2.2 Database Schema & Domain Entity Relationships - 2.3 Protocols & Communications (REST, MQTT, AI)

Chapter 3: Consumer (Receiver) Portal Operating Manual

- 3.1 Live Energy Dashboard ([OverviewScreen.tsx](#)) - 3.2 Smart Meter Onboarding & Invites Hub ([ConsumerMeterRegistrationScreen.tsx](#)) - 3.3 Prepaid Utility Wallet & Invoicing ([BillingScreen.tsx](#)) - 3.4 P2P Energy Marketplace & Escrow ([TradingScreen.tsx](#)) - 3.5 AI Assistant & Energy Forecasting ([AIChatScreen.tsx](#) & [AIForecastingScreen.tsx](#)) - 3.6 Alarm Center & System Notifications ([AlarmScreen.tsx](#)) - 3.7 Account Profile & Security Controls ([SettingsScreen.tsx](#))

Chapter 4: Producer & Prosumer Portal Operating Manual

- 4.1 Master Producer Overview ([ProducerDashboardScreen.tsx](#)) - 4.2 Consumer Connection & Load Dispatch Hub ([ProducerConsumersScreen.tsx](#)) - 4.3 Commissioning & Setup Wizard ([ProducerSetupWizardScreen.tsx](#)) - 4.4 Microgrid Busbar & Load Balancing Topology ([ProducerMicrogridScreen.tsx](#)) - 4.5 Fleet Portfolio Monitoring ([FleetScreen.tsx](#)) - 4.6 Producer

Billing & Financial Center ([ProducerBillingScreen.tsx](#) & [AdminScreen.tsx](#)) - 4.7 Engineering Solar Design & Cable Sizing Tool ([SolarDesignScreen.tsx](#))

Chapter 5: Producer <-> Consumer Connection Protocol

- 5.1 Protocol Lifecycle & State Transition Matrix - 5.2 Sequence Diagram: Consumer Invite -> Producer Acceptance - 5.3 Exception Handling & Automated Fail-Safes

Chapter 6: Developer & Operations Guide

- 6.1 Prerequisites & Local Installation - 6.2 Environment Configurations ([.env](#)) - 6.3 Swagger API Reference Summary - 6.4 Troubleshooting & Maintenance Matrix

CHAPTER 1: INTRODUCTION & CORE SYSTEM PHILOSOPHY

1.1 Executive Vision & Product Scope

REOS (Renewable Energy Operating System) is an integrated software platform engineered to orchestrate modern clean energy microgrids, commercial solar power stations, prosumer micro-generators, and residential/commercial energy consumers.

Modern renewable energy grids face a fundamental challenge: bridging the physical generation of clean power with real-time IoT telemetry, automated grid dispatching, and friction-free financial settlements. REOS provides a unified solution that transforms static solar installations into dynamic, commercial-grade energy utilities.

1.2 Target Personas & Use Cases

PersonaRoleKey Objectives in REOS

Energy Consumer (Receiver)Residential home or commercial business receiving powerBind smart meters, track prepaid wallet balance, request power allocations from local producers, participate in P2P energy trading. **Solar Prosumer**Residential or commercial owner with solar PV + batteryExport surplus power to neighbors or microgrid, set custom tariffs, manage incoming receiver requests. **Commercial Plant Producer**Industrial Solar Farm / IPP operator (100kW – 10MW+)Onboard utility plants, balance feeder zones, manage consumer connections via **Receiver Smart Meter Numbers**, monitor grid stability. **System Administrator**Grid regulator or financial controllerAudit platform revenue streams, inspect escrow transactions, oversee system health and hardware diagnostics.

CHAPTER 2: ARCHITECTURE & TECHNICAL FOUNDATIONS

2.1 Full-Stack Monorepo Structure

REOS is organized as a clean TypeScript monorepo combining a cross-platform React Native frontend with a NestJS backend:

```
new project/
├── apps/
│   ├── frontend/                                # React Native / Expo Application
│   │   ├── src/
│   │   │   ├── navigation/
│   │   │   │   └── AppNavigator.tsx            # Drawer & Native Stack Navigation Registry
│   │   │   ├── screens/                        # 25 Dedicated Screen Interfaces
│   │   │   └── store/
│   │   │       ├── useStore.ts                # Zustand Global Store & Offline Fallbacks
│   │   │       └── api.ts                     # Axios/Fetch HTTP Client
│   │   └── App.tsx                            # Main App Bootstrapper
│   └── backend/                                # NestJS Backend API Core
│       ├── src/
│       │   ├── devices/                       # IoT Device & Onboarding Controller/Service
│       │   ├── consumer/                      # Consumer Contract & Wallet Logic
│       │   ├── producer/                     # Plant Management & Dispatch Engine
│       │   ├── trading/                      # P2P Order Book & Escrow Manager
│       │   ├── ai/                           # Google Gemini AI Integration Engine
│       │   └── prisma/                        # Database Schema & Migrations
│       └── main.ts                            # NestJS Entry Point
```

2.2 Database Schema & Domain Entities

REOS relies on Prisma ORM backing a PostgreSQL relational schema. Key domain models include:

User: Stores credentials, role (`PROSUMER` , `CONSUMER` , `PRODUCER` , `ADMIN`), contact info, and security tokens.

Device: Physical IoT hardware registered in the system (Type: `INVERTER` , `SMART_METER` , `BATTERY` , `GATEWAY`). Includes serial numbers, MQTT topics, and verification status.

SmartMeter: Specific profile linked to a `Device` record containing phase configuration (`SINGLE_PHASE` / `THREE_PHASE`), meter number, and installation address.

Plant: Producer utility station profile (Capacity kW, plant type, feeder zones, master smart meters).

ConsumerConnection: Junction entity mapping a Consumer's Smart Meter to a Producer Plant. Stores status ([PENDING](#) , [CONNECTED](#) , [REJECTED](#) , [SUSPENDED](#)), allocated power (kW), and request messages.

Wallet: Prepaid utility currency account (Currency: [NGN](#)), available balance, locked escrow balance, and billing cycle.

Transaction & Invoice: Audit ledger recording wallet top-ups, energy bill deductions, P2P trade settlements, and monthly statement invoices.

TradeOffer / BuyRequest / Escrow: P2P energy trading order book and escrow contract entities.

CHAPTER 3: CONSUMER (RECEIVER) PORTAL OPERATING MANUAL

The Consumer Portal empowers energy receivers to monitor power draw, manage utility payments, and connect with clean energy producers.

```
+-----+
CONSUMER DASHBOARD
[ ⚡ Prepaid Utility Wallet: ₦5,000.00 ] --> [ 💵 Top Up Wallet ]

[ 🏠 Register Smart Meter ]           [ 🔌 Request Connection ]

[ Usage KPIs ]
🏠 Current Load: 1.25 kW           📊 Daily Usage: 8.4 kWh
🔄 Microgrid Recv: 6.2 kWh         🔌 Grid Import: 2.2 kWh

[ Live Energy Flow Path ]
+-----+
```

3.1 Live Energy Dashboard ([OverviewScreen.tsx](#))

Wallet Card & Top-Up Trigger:

Balance Display: Live prepaid energy balance (e.g. [₦5,000.00](#)).

Top Up Wallet Button: Opens the **Interactive Top-Up Modal**. Consumers select quick amount presets ([₦1,000](#) , [₦2,500](#) , [₦5,000](#) , [₦10,000](#) , [₦25,000](#)) or custom amounts, pick a payment gateway (*Paystack, Flutterwave, Direct Bank*), and receive immediate account credit with visual feedback.

Action Hub:

Register Smart Meter Button: Triggers the interactive Smart Meter Registration modal and navigates to the registration hub.

Request Connection Button: Opens the Producer Connection Request modal to send a dispatch invite.

Link Supplier Input: Enter a 6-digit Supplier Invitation Code (e.g., `REOS-1234`) to bind directly with a local microgrid.

Interactive Usage Details (KPI Cards):

Tapping any KPI card opens an **Interactive Metric Modal**:

- **Current Load:** Shows active kW draw, voltage (231V), current (5.4A), and line frequency (50.0Hz). -

Daily Usage: Cumulative kWh consumed today and peak usage timeframes. - **Microgrid Recv.:** Clean solar energy delivered from linked suppliers at the agreed tariff (e.g., `#180/kWh`). - **Grid Import:** Utility backup power imported during non-solar hours. - **Est. Cost Today:** Daily expenditure breakdown and total financial savings versus utility grid tariffs.

Interactive Smart Flow Diagram:

Tap the flow nodes (*Microgrid Supplier, Smart Meter, Home Load*) to inspect live MQTT telemetry streams, hardware ping latency, and power factor diagnostics.

3.2 Smart Meter Onboarding & Invites Hub

(`ConsumerMeterRegistrationScreen.tsx`)

```

+-----+
SMART METER REGISTRATION & INVITES

1. Meter Specification Form
- Meter Name: [ Home Main Smart Meter ]
- Serial / Utility ID: [ CNS-MTR-778899 ]
- Phase: (x) Single Phase ( ) Three Phase
[ Register & Bind Smart Meter ]

2. Registered Smart Meters List & Connectivity Test
🏠 Ade Residential Smart Meter (SN: CNS-MTR-778899) • VERIFIED
[ 🔄 Run Connectivity Verification Test ]

3. Energy Connection Invites Sent & Live Tracker
🏢 Alpha Commercial Solar Farm (100kW)
Receiver Meter: CNS-MTR-778899 • Requested: 5.0 kW
Status: [ ⌚ PENDING PRODUCER APPROVAL ]
+-----+

```

Step-by-Step Operating Instructions:

Register Physical Smart Meter:

- Fill in Meter Name, Serial Number (e.g., [CNS-MTR-778899](#)), Utility ID, Manufacturer, and Phase Configuration ([SINGLE_PHASE](#) / [THREE_PHASE](#)). - Click **Register & Bind Smart Meter**. The meter is assigned to your authenticated user account.

Run Connectivity Diagnostic Test:

- Tap **Run Connectivity Verification Test** on any registered meter. - The system executes a ping check against the MQTT telemetry broker and verifies latency (e.g. 12ms) and signal quality (94%).

Send Connection Invite to Energy Producer:

- Search available Producer Plants & Prosumer Microgrids by name or ID. - Click **Submit Connection Request**. Specify requested power allocation (e.g., [5.0 kW](#)) and custom note.

Track Invite Status:

- View your invite under **Connection Invites Sent & Live Status Tracking**. - Live Status Indicators: - [⌚ PENDING](#) : Invite sent to Producer; awaiting approval on Producer Dashboard. - [✅ CONNECTED](#) : Producer accepted invite. Automated power dispatch and net billing active. - [❌ REJECTED](#) : Invite declined by Producer.

3.3 Prepaid Utility Wallet & Invoicing ([BillingScreen.tsx](#))

Utility Wallet Management: Check available funds, last payment timestamp, and prepaid tariff conditions.

Invoice Statement Ledger: Review itemized monthly energy invoices detailing total kWh received, rate per kWh, tax breakdown, and due dates.

Pay Outstanding Invoice: One-click payment for unpaid or overdue invoices using utility wallet balance.

Transaction History: Audit trail of all prepaid purchases, trade settlements, and P2P transfers.

3.4 P2P Energy Marketplace & Escrow ([TradingScreen.tsx](#))

REOS features a Peer-to-Peer Energy Marketplace allowing consumers and prosumers to buy and sell surplus solar energy.

Market Order Book: Browse open sell offers and buy requests with location tags and price per kWh ([¥185 - ¥220/kWh](#)).

Instant Buy & Trade: Click **Buy Energy** on any offer. The system calculates energy cost + platform fee ([¥500](#)) and locks funds in Escrow.

Escrow Settlement: Upon meter verification of energy transfer, funds are released to the seller.

Dispute Center: Raise a formal dispute if energy delivery fails. Administrators review telemetry logs to refund or settle escrows.

3.5 AI Assistant & Energy Forecasting ([AIChatScreen.tsx](#) & [AIForecastingScreen.tsx](#))

AI Energy Assistant ([AIChatScreen.tsx](#)): Powered by Google Gemini. Ask questions like: "*How can I optimize my battery usage during peak hours?*" or "*What is my average daily expenditure?*".

AI Forecasting ([AIForecastingScreen.tsx](#)): Predicts 7-day, 30-day, and 90-day solar generation and household load demand patterns.

CHAPTER 4: PLANT PRODUCER & PROSUMER PORTAL OPERATING MANUAL

Designed for Commercial Solar Plants (100kW – 10MW+) and Prosumers managing energy distribution to connected consumers.

```
+-----+
PRODUCER CONSUMER MANAGEMENT
Tabs: [ Connected Consumers (4) ]   [ Invites Requests (1) ]

INCOMING CONSUMER CONNECTION INVITES
👤 Ade Okafor (ade.okafor@example.com) • PENDING
📡 Receiver Smart Meter Number: [ CNS-MTR-778899 ]
Requested Power Allocation: 5.0 kW
Note: "Requesting clean solar energy allocation for home."

[ ✅ Accept Invite & Connect Receiver ]   [ Decline ]
+-----+
```

4.1 Master Producer Overview ([ProducerDashboardScreen.tsx](#))

Capacity Overview: Displays installed solar capacity (kW), live generation output (kW), active connected consumers count, and daily revenue generation (₦).

Master Grid Control Switch: Toggle power dispatch on/off across all connected feeders.

Generation vs. Demand Gauge: Visual ratio of live solar production versus connected consumer load demand.

4.2 Consumer Connection & Load Dispatch Hub

([ProducerConsumersScreen.tsx](#))

This screen represents the Producer-side interface for managing receiver connections.

Incoming Consumer Connection Invites Tab:

Review pending connection invites submitted by consumers.

Inspect the **Receiver Smart Meter Number** (e.g. [CNS-MTR-778899](#)), consumer name, email, phone number, and requested kW power allocation.

Click [\[✅ Accept Invite & Connect Receiver \]](#) :

- The connection status updates to [CONNECTED](#) . - The consumer's receiver meter is added to the active dispatch list. - The consumer's dashboard automatically reflects the [CONNECTED](#) state.

Click [\[Decline\]](#) to reject an unsuitable connection request.

Connected Consumers Tab:

Monitor live power draw (kW) for each connected consumer receiver meter.

View remaining energy allocation balances (kWh).

Toggle individual load switches:  **Disconnect Load** or  **Connect Load**.

4.3 Setup & Commissioning Wizard ([ProducerSetupWizardScreen.tsx](#))

A 4-step wizard for onboarding new commercial energy infrastructure:

Plant Details: Name, location, organization, and total capacity (kW).

Master Producer Smart Meter: Serial number, meter number (e.g. [PRD-MTR-001](#)), and MQTT gateway topic.

Inverters & PV Arrays: Onboard inverter specifications, efficiency ratings, and string configurations.

Battery Energy Storage System (BESS): Capacity (kWh), battery voltage (V), and maximum discharge current.

4.4 Microgrid Topology & Load Balancing

([ProducerMicrogridScreen.tsx](#))

Live Flow Path Diagram: Visualizes energy movement from Solar PV -> Battery Storage -> Consumer Load -> Utility Grid Net Export.

Busbar Phase Voltage Monitoring: Displays Phase A, B, and C AC voltages (e.g. 231.2V) to ensure grid compliance.

Line Loss Calculator: Calculates distribution cable heat losses and percentage loss efficiency.

4.5 Engineering Tools ([SolarDesignScreen.tsx](#))

Includes interactive engineering calculators for installers and plant managers:

Solar PV Sizing Tool: Enter daily energy target (kWh) to calculate required solar panel wattage and roof footprint.

Battery Backup Calculator: Compute required amp-hour (Ah) battery bank size based on autonomy days and depth of discharge (DoD).

Cable Voltage Drop Calculator: Input current (A), cable distance (meters), voltage (V), and wire cross-sectional area (mm²) to compute voltage drop percentage.

CHAPTER 5: PRODUCER <-> CONSUMER CONNECTION PROTOCOL

5.1 Protocol State Transition Matrix

Every connection request follows a strict state machine to prevent unauthorized energy draw:

Initial StateTrigger ActionNew StateDescription

NONE Consumer submits invite on **ConsumerMeterRegistrationScreen** **PENDING** Connection invite created with Receiver Smart Meter Number (**CNS-MTR-XXXXXX**) and requested kW allocation.

PENDING Producer clicks **Accept Invite** on **ProducerConsumersScreen** **CONNECTED** Producer binds receiver meter. Automated power dispatch and net billing activated.

PENDING Producer clicks **Decline** **REJECTED** Request rejected. Consumer notified on dashboard tracker.

CONNECTED Producer toggles **Disconnect Load** **SUSPENDEED** / **DISCONNECTED** Power dispatch physically isolated. Telemetry continues logging standby state.

SUSPENDEED Producer toggles **Connect Load** **CONNECTED** Power dispatch restored to receiver meter.

5.2 End-to-End Sequence Diagram

```

mermaid
sequenceDiagram
    autonumber
    actor Consumer
    participant ConsumerUI as Consumer Meter Screen
    participant Store as Zustand Store / API
    participant ProducerUI as Producer Dashboard
    actor Producer

    Consumer->>ConsumerUI: 1. Fill Meter Specs (CNS-MTR-778899) & Click Register
    ConsumerUI->>Store: registerConsumerSmartMeter(dto)
    Store-->>ConsumerUI: Meter Bound to Consumer Account

    Consumer->>ConsumerUI: 2. Search Producer Plant & Click "Send Invite"
    ConsumerUI->>Store: submitConnectionRequest({ plantId, requestedKw: 5.0 })
    Store-->>ConsumerUI: Status set to PENDING

    Note over ConsumerUI: Step 4 Tracker shows:
    " ⌚ PENDING PRODUCER APPROVAL "

    Producer->>ProducerUI: 3. Open "Incoming Invites" tab
    ProducerUI-->>Producer: Card displays:
    Receiver Meter: CNS-MTR-778899
    Requested Power: 5.0 kW

    Producer->>ProducerUI: 4. Click "[✅ Accept Invite & Connect Receiver]"
    ProducerUI->>Store: processConnectionApproval({ connectionId, action: 'APPROVE' })
    Store-->>ProducerUI: Status updated to CONNECTED

    Note over ConsumerUI: Step 4 Tracker automatically updates to:
    " ✅ CONNECTED (Energy Dispatch Active) "

```

CHAPTER 6: DEVELOPER & OPERATIONS GUIDE

6.1 Prerequisites & Installation

Environment Requirements

Node.js v18.0.0 or higher

npm v9.0.0 or higher

Git

Installation Steps

```
bash
```

```
git clone https://github.com/uzo-john/REOS.git  
cd REOS
```

```
cd apps/frontend  
npm install
```

```
cd ../backend  
npm install
```

6.2 Running the Application

Frontend (React Native / Expo Web)

```
bash  
cd apps/frontend
```

```
npx expo start --web
```

Backend (NestJS Server)

```
bash
cd apps/backend
```

```
npx prisma generate
```

```
npm run start:dev
```

Access API Documentation (Swagger): <http://localhost:3000/api/docs>

6.3 Key REST API Endpoints Summary

EndpointMethodRoleDescription

[/api/devices/consumer/register-smart-meter](#) POST CONSUMER Registers physical smart meter serial number and profile. [/api/devices/consumer/submit-connection-request](#) POST CONSUMER Sends connection invite from smart meter to target producer plant. [/api/devices/producer/connection-requests](#) GET PRODUCER Returns incoming consumer invites containing receiver smart meter numbers. [/api/devices/producer/process-approval](#) POST PRODUCER Approves ([CONNECTED](#)) or rejects ([REJECTED](#)) consumer connection requests. [/api/wallets/recharge](#) POST CONSUMER Credits prepaid utility wallet via simulated payment gateway. [/api/trading/buy-energy](#) POST CONSUMER Initiates P2P trade purchase and locks funds in escrow.

6.4 Troubleshooting & Support Matrix

IssueRoot CauseResolution

Button appears static on web browserMissing [activeOpacity](#) or click event handlerUpdate component with [. Connection invite not showing on Producer screenPlant ID mismatch or un-fetched request listTrigger](#) [fetchProducerConnectionRequests\(\)](#) in [useEffect](#) when switching tabs. Wallet top-up

balance not updatingAsynchronous state delayCall `await fetchConsumerBilling()` immediately following `rechargeWallet()` . Smart Meter connectivity test failsMQTT broker offline or serial number unregisteredVerify serial number format (`CNS-MTR-XXXXXX`) and run diagnostic ping test.

REOS Complete User Operating Manual & Technical System Specification — Version 2.0